

Data-Driven Recursive Least Squares Estimation for Model Predictive Current Control of Permanent Magnet Synchronous Motors

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Abstract—The performance of model predictive controllers (MPC) strongly depends on the quality of their models. In the field of electric drive control, white-box (WB) modeling approaches derived from first-order physical principles are most common. This procedure typically does not cover parasitic effects and often comes with parameter deviations. These issues are particularly crucial in the domain of self-commissioning drives where a hand-tailored, accurate WB plant model is not available. In order to compensate for such modeling errors and, consequently, to improve the control performance during transients and steady state, this article proposes a data-driven, real-time capable recursive least squares estimation method for the current control of a permanent magnet synchronous motor. Following this machine learning approach, the effect of the flux linkage and voltage harmonics due to the winding scheme can also be taken into account through suitable feature engineering. Moreover, a compensating scheme for the interlocking time of the inverter is proposed. The resulting algorithm is investigated using the well-known finite-control-set MPC (FCS-MPC) in the rotor-oriented coordinate system. The extensive experimental results show the superior performance of the presented scheme compared to a FCS-MPC-based on a state-of-the-art WB motor model using look-up tables for addressing (cross-)saturation.

Index Terms—Finite-control-set, identification, model predictive control, permanent magnet synchronous motor (PMSM), recursive least squares (RLS), self-commissioning, machine learning.

I. INTRODUCTION

DUE to the increasing computing power, model predictive control (MPC) is gaining popularity in the field of power electronics and electrical drives [1]–[5]. In the context of field-oriented motor control, the current controller represents the important basis of all further control loops. The main requirements for the current controller of an electrical drive is a zero steady state error and—in transient operation—a short settling time with little overshoot. Also secondary requirements such as low current distortion can be formulated.

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The performance of MPC depends largely on the accuracy of a given system's model. For current control of a permanent magnet synchronous motor (PMSM), the basic fundamental-wave motor model [3] with fixed motor parameters is usually used. Relying solely on such a model can lead to large prediction errors and, thus, to a worse performance compared to other control techniques such as linear field-oriented approaches based on proportional-integral feedback. Prediction errors due to model deviations can be caused by several reasons such as varying motor parameters, inverter nonlinearities, and flux linkage harmonics. In general, motor parameters such as inductances, ohmic resistances, etc. depend on temperature, frequency, and (cross-)saturation effects [6], [7], as well as production deviations and aging processes [8], [9]. A possible solution to overcome the problem of model deviation due to varying motor parameters is to identify them for all occurring operating conditions and store them in look-up tables (LUT) [10]. However, due to the large amount of different root causes it is unrealistic to cover all of them in offline identified LUTs, especially when they originate from motor-specific influences (e.g., production deviations) or are induced over time (e.g., material aging). Another approach is to identify the motor parameters online. Frequently used methods are Kalman-filter-based identification algorithms [11], [12]. Here, a white-box (WB) model with physical parameters is usually utilized, and only a selected subset of those parameters is identified online. In the context of sensorless control, a broad range of recursive least squares (RLS) methods for estimating a WB model exists [13]–[15]. There, the varying rotor speed and position information is extracted from a WB model, which is typically designed by an expert engineer.

Instead of identifying the motor parameters, disturbance variables can also be introduced into the model description and can be identified as well as compensated online with suitable observer methods to improve the robustness of the current controller [16]. Different observer concepts such as extended Kalman filter [17], sliding-mode observer [18] or moving-horizon estimator [19] can be used here. Lyapunov-based methods can further increase or even guarantee robustness with additional constraints in the optimization problem of the finite-control-set MPC (FCS-MPC) [20], [21]. Nevertheless, it should be noted that guaranteed robustness does not imply an optimal control performance. Hence, it is always favorable to eliminate model deviations as the root cause requiring robust controller designs.

In contrast, the data-driven method presented here does not use a predefined model structure consisting of physical parameters. As a result, the data-driven approach allows more flexibility than all mentioned WB-based models and, therefore, can adapt better to the present operation condition. The estimator merely identifies the behavior between inputs and outputs in a data-driven fashion. Model-free approaches can also be interpreted as data-driven methods [22]–[25]. These methods do not require any physical parameters. Here, the current variation between two sampling instants, belonging to the corresponding elementary voltage vector, is stored in a LUT. These current variations are used for current prediction and the voltage vector that leads to the minimal control error is selected. Although model-free methods are an effective and simple method for FCS-MPC, their performance can be critical for high speeds or low pulse numbers because of their inherent working principle. Also, the papers available on model-free approaches are tested only on low power and low utilize factor motors, which behavior can be easily modeled in a linear time-invariant fashion. Hence, it remains unclear if those approaches are also capable handling more complex motors such as highly utilized traction drives.

In [26], a RLS method for predictive current control of a PMSM is described. Here, only two regressors were used respectively for the q - and d -axis and the nonideal inverter behavior was neglected. A least squares (LS) method to estimate the motor model for a PMSM, also in a data-driven fashion and in the context of model predictive control can be found in [27]. This method estimates one autonomous system for every switching state. After recording a series of measurements online, the LS method estimates the autonomous systems and update the model to improve the performance of the controller at the present operating point. However, the motor model is not recursively updated after each controller iteration. In comparison to [27], the motor model presented here is continuously estimated online using a RLS method, and an inverter nonlinearity compensation is taken into account.

Techniques to compensate for nonlinearities of inverters are described in [14], [28]–[30], which all require a modulator. Since a modulator is not required when using FCS-MPC, these compensation techniques drop out of consideration upfront. Instead, a compensation scheme for the interlocking time of the inverter respecting the FCS-MPC properties is presented.

The well-known FCS-MPC with a prediction horizon of $N = 1$ is a suitable choice for the current control loop because of its low computational complexity and the short settling time with overshoot-free response [31]. To compensate for the computation time delay of the FCS-MPC, an additional prediction step at the beginning of each controller iteration is executed [32]. A further advantage of this control method, compared to modulator-based techniques (e.g., continuous-control-set MPC) is the inherent system excitation of the applied switching sequence, which is beneficial for any online identification. Moreover, the proposed identification is independent from the induced voltage in the fundamental-wave domain, and, therefore, can be used in the entire speed range including standstill. Compared to any other indirect control approach utilizing, a modulator with regular sampling, no additional signal injection is required

for the proposed control and identification scheme making it particularly suitable for self-commissioning drive applications.

II. MATHEMATICAL MODEL

The following equations summarize the state-of-the-art WB motor model, the effect of the inverter interlocking time and the FCS-MPC.

A. Coordinate Systems

Transformations between stator-fixed three-phase abc and the stator-fixed $\alpha\beta$ coordinate system can be calculated with the following matrices:

$$\mathbf{T}_{\alpha\beta abc} = \frac{2}{3} \begin{bmatrix} 1 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & \frac{\sqrt{3}}{2} & -\frac{\sqrt{3}}{2} \end{bmatrix}, \quad \mathbf{T}_{abc\alpha\beta} = \mathbf{T}_{\alpha\beta abc}^\dagger. \quad (1)$$

Here, $\dagger$ denotes the Moore–Penrose pseudoinverse and bold symbols depict matrices/vectors. In a similar way, the transformations between the $\alpha\beta$ and rotor-fixed dq coordinate system can be formulated

$$\mathbf{T}_{dq\alpha\beta}(\varepsilon(t)) = \begin{bmatrix} \cos(\varepsilon(t)) & \sin(\varepsilon(t)) \\ -\sin(\varepsilon(t)) & \cos(\varepsilon(t)) \end{bmatrix} \quad (2)$$

$$\mathbf{T}_{\alpha\beta dq}(\varepsilon(t)) = \mathbf{T}_{dq\alpha\beta}^{-1}(\varepsilon(t))$$

in which ε denotes the electrical rotation angle of the PMSM. With (1) and (2), the transformation between the abc and dq coordinate system can be obtained

$$\mathbf{T}_{dqabc}(\varepsilon(t)) = \mathbf{T}_{dq\alpha\beta}(\varepsilon(t))\mathbf{T}_{\alpha\beta abc}$$

$$\mathbf{T}_{abcdq}(\varepsilon(t)) = \mathbf{T}_{dqabc}^\dagger(\varepsilon(t)). \quad (3)$$

B. Discrete-Time PMSM WB Model

Starting from a general motor model, a simplified motor model is derived to discuss the influence of (cross-)saturation effects and small-angle approximation on the density/sparsity of the state-space system matrices.

1) *Generalized Fundamental-Wave Model*: The current-based discrete-time difference equation of a PMSM considering (cross-)saturation effects in the stator-based dq coordinate system can be described as follows [33]:

$$\begin{aligned} \mathbf{i}_{dq}[k+1] &= \left(\mathbf{I} - \mathbf{L}_{dq,\Delta}^{-1} R_s T_s \right) \mathbf{i}_{dq}[k] \\ &+ \mathbf{L}_{dq,\Delta}^{-1} \mathbf{T}_{\alpha\beta dq}(-T_s \omega) T_s \mathbf{u}_{dq}[k] \\ &+ \mathbf{L}_{dq,\Delta}^{-1} (\mathbf{T}_{\alpha\beta dq}(-T_s \omega) - \mathbf{I}) \boldsymbol{\psi}_{dq}[k] \end{aligned} \quad (4)$$

where $\mathbf{i}_{dq} = [i_d \ i_q]^\top$ is the current, $\mathbf{u}_{dq} = [u_d \ u_q]^\top$ the stator voltage, $\boldsymbol{\psi}_{dq} = [\psi_d \ \psi_q]^\top$ the flux linkage, R_s the stator resistance, ω the electrical angular velocity, T_s the constant time interval between two samples, $\mathbf{I}$ the unit matrix, and $\mathbf{L}_{dq,\Delta}$ the differential inductance matrix defined by

$$\mathbf{L}_{dq,\Delta}(\mathbf{i}_{dq}, \omega)[k] = \begin{bmatrix} L_{dd}[k] & L_{dq}[k] \\ L_{qd}[k] & L_{qq}[k] \end{bmatrix}. \quad (5)$$

The differential inductance matrix is a function of the current and electric angular velocity because of (cross-)saturation effects and iron losses, respectively. Due to the same effects, the flux is also a nonlinear function of the current and the electric angular velocity $\psi_{dq} = \psi_{dq}(i_{dq}, \omega)$. The discrete-time difference equation (4) can be formulated in state-space representation

$$\mathbf{i}_{dq}[k+1] = \mathbf{A}[k]\mathbf{i}_{dq}[k] + \mathbf{B}[k]\mathbf{u}_{dq}[k] + \mathbf{E}[k]$$

$$\text{with: } \mathbf{A}[k] = \left(\mathbf{I} - \mathbf{L}_{dq,\Delta}^{-1} R_s T_s \right) = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

$$a_{11} = 1 - \frac{L_{qq}}{|\mathbf{L}_{dq,\Delta}|} R_s T_s, \quad a_{12} = \frac{L_{dq}}{|\mathbf{L}_{dq,\Delta}|} R_s T_s$$

$$a_{21} = \frac{L_{qd}}{|\mathbf{L}_{dq,\Delta}|} R_s T_s, \quad a_{22} = 1 - \frac{L_{dd}}{|\mathbf{L}_{dq,\Delta}|} R_s T_s$$

$$\mathbf{B}[k] = \mathbf{L}_{dq,\Delta}^{-1} \mathbf{T}_{\alpha\beta dq}(-T_s \omega) T_s = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix}$$

$$b_{11} = \frac{L_{qq} \cos(-T_s \omega) - L_{dq} \sin(-T_s \omega)}{|\mathbf{L}_{dq,\Delta}|} T_s$$

$$b_{12} = -\frac{L_{dq} \cos(-T_s \omega) + L_{qq} \sin(-T_s \omega)}{|\mathbf{L}_{dq,\Delta}|} T_s$$

$$b_{21} = -\frac{L_{qd} \cos(-T_s \omega) - L_{dd} \sin(-T_s \omega)}{|\mathbf{L}_{dq,\Delta}|} T_s$$

$$b_{22} = \frac{L_{dd} \cos(-T_s \omega) + L_{qd} \sin(-T_s \omega)}{|\mathbf{L}_{dq,\Delta}|} T_s$$

$$\mathbf{E}[k] = \mathbf{L}_{dq,\Delta}^{-1} (\mathbf{T}_{\alpha\beta dq}(-T_s \omega) - \mathbf{I}) \psi_{dq}[k] = \begin{bmatrix} e_1 \\ e_2 \end{bmatrix}$$

$$e_1 = -\frac{(L_{dq} \sin(-T_s \omega) - L_{qq}(\cos(-T_s \omega) - 1))\psi_d}{|\mathbf{L}_{dq,\Delta}|}$$

$$-\frac{(L_{qq} \sin(-T_s \omega) + L_{dq}(\cos(-T_s \omega) - 1))\psi_q}{|\mathbf{L}_{dq,\Delta}|}$$

$$e_2 = \frac{(L_{dd} \sin(-T_s \omega) - L_{qd}(\cos(-T_s \omega) - 1))\psi_d}{|\mathbf{L}_{dq,\Delta}|}$$

$$\frac{(L_{qd} \sin(-T_s \omega) + L_{dd}(\cos(-T_s \omega) - 1))\psi_q}{|\mathbf{L}_{dq,\Delta}|}.$$

(6)

Above, it can be seen that the system matrices $\mathbf{A}[k]$, $\mathbf{B}[k]$, and $\mathbf{E}[k]$ are dense in general.

2) *Simplified Fundamental-Wave Model:* If (cross-) saturation effects and iron losses are neglected the flux ψ_{dq} can be expressed with the absolute inductance matrix $\mathbf{L}_{dq}$ and the permanent magnet flux ψ_p

$$\psi_{dq} = \mathbf{L}_{dq} \mathbf{i}_{dq} + \psi_p = \begin{bmatrix} L_d & 0 \\ 0 & L_q \end{bmatrix} \mathbf{i}_{dq} + \begin{bmatrix} \psi_p \\ 0 \end{bmatrix}. \quad (7)$$

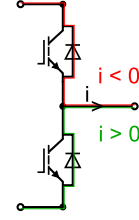


Fig. 1. In this definition, a positive current pertains to an active lower with an inactive upper switch and negative current vice versa during the interlocking time.

In addition to that, the differential inductance matrix is identical with the absolute inductance matrix, such that $\mathbf{L}_{dq,\Delta} = \mathbf{L}_{dq}$. If a small-angle approximation (small change of angle $\Delta\varepsilon = T_s \omega$ per sample time T_s , such that $\cos(\Delta\varepsilon) \approx 1$ and $\sin(\Delta\varepsilon) \approx \Delta\varepsilon$) is also applied, the generalized fundamental-wave model (6) simplifies to the commonly used discrete state-space representation of the PMSM with the system matrices

$$\mathbf{A} = \begin{bmatrix} 1 - \frac{R_s T_s}{L_d} & \frac{L_q \omega T_s}{L_d} \\ -\frac{L_d \omega T_s}{L_q} & 1 - \frac{R_s T_s}{L_q} \end{bmatrix} \quad (8)$$

$$\mathbf{B} = \begin{bmatrix} \frac{T_s}{L_d} & 0 \\ 0 & \frac{T_s}{L_q} \end{bmatrix}, \quad \mathbf{E} = \begin{bmatrix} 0 \\ -\frac{\psi_p \omega T_s}{L_q} \end{bmatrix}$$

In comparison to the generalized fundamental-wave model (6), the system matrices $\mathbf{B}$ and $\mathbf{E}$ of the simplified fundamental-wave model (8) are sparse.

C. Inverter

The finite set of the voltages in the $\alpha\beta$ coordinate system due to the finite number of possible switching combinations of the three-phase two-level inverter can be defined as

$$\mathbf{u}_{\alpha\beta} \in \{\mathbf{u}_{\alpha\beta} \in \mathbb{R}^2 | \mathbf{u}_{\alpha\beta} = u_{DC} \mathbf{T}_{\alpha\beta abc} \mathbf{s}_{abc}\} \quad (9)$$

where u_{DC} is the dc-link voltage and $\mathbf{s}_{abc}$ is the switching state of the inverter defined by

$$\mathbf{s}_{abc} = [s_a \quad s_b \quad s_c]^T \in \{0, 1\}^3 \subset \mathbb{Z}^3. \quad (10)$$

With the equations above, the stator voltage $\mathbf{u}_{dq}$ can be calculated

$$\mathbf{u}_{dq} = \mathbf{T}_{dq\alpha\beta}(\varepsilon) \mathbf{u}_{\alpha\beta} = u_{DC} \mathbf{T}_{dq\alpha\beta}(\varepsilon) \mathbf{s}_{abc}. \quad (11)$$

To avoid a hard short circuit in the inverter, switching is delayed for the interlocking time T_i in case the other transistor in a phase leg was previously in conducting mode. During this time, both transistors are OFF and the phase current flows through one of the two diodes. Depending on the sign of the current, the phase of the PMSM is then connected to the upper or lower dc-link potential via the conducting diode (see Fig. 1). In the following, the resulting switching state shall be denoted as intermediate

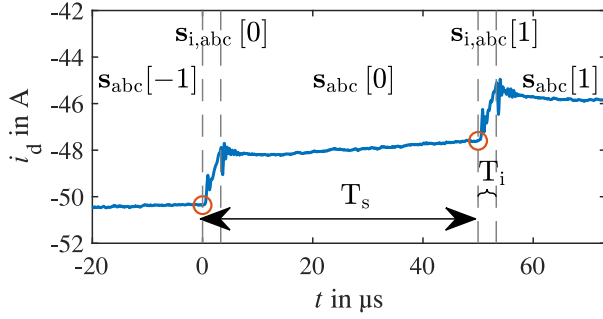


Fig. 2. Exemplary trajectory of the i_d current, where the effect of the intermediate vectors $\mathbf{s}_{i,abc}[0]$ and $\mathbf{s}_{i,abc}[1]$ during the interlocking time T_i can be seen. Also, the sample points (red) and the sample time T_s of the controller is depicted.

switching vector $\mathbf{s}_{i,abc}$ (see Fig. 2). The number of zero crossings of a phase current during the interlocking time is assumed to be negligible small. The intermediate switching vector $\mathbf{s}_{i,abc}$ at the k th time instant without discontinuous conduction can be computed as follows:

$$\mathbf{s}_{i,abc}[k] = |\mathbf{s}_{abc}[k] - \mathbf{s}_{abc}[k-1]| \odot h(-\mathbf{i}_{abc}[k]) + ||\mathbf{s}_{abc}[k] - \mathbf{s}_{abc}[k-1]| - \mathbf{1}| \odot \mathbf{s}_{abc}[k] \quad (12)$$

with the heaviside function h , the elementwise multiplication $\odot$ and $\mathbf{1} = [1 \ 1 \ 1]^T$. Usually, the effect of the interlocking time is neglected for FCS-MPC, which leads to a systematic modeling error and, consequently control error.

D. Finite-Control-Set Model Predictive Control

After the calculation of the intermediate vector has been shown, it can be integrated into the prediction and thus into the optimization problem of the FCS-MPC. The cost function for the FCS-MPC with prediction horizon of $N = 1$ is designed to penalize the squared Euclidean distance between the predicted current of the next time step and the reference current $\mathbf{i}_{dq}^*$. With the definition of the cost function the optimization problem can be formulated as

$$\min_{\mathbf{s}_{abc}[k]} \|\mathbf{i}_{dq}[k+1] - \mathbf{i}_{dq}^*[k+1]\|_2^2$$

$$\text{s.t.} \quad \mathbf{u}_{dq}[k] = u_{DC} \mathbf{T}_{dqabc}(\varepsilon[k]) \mathbf{s}_{abc}[k] \quad (13a)$$

$$\mathbf{u}_{i,dq}[k] = u_{DC} \mathbf{T}_{dqabc}(\varepsilon[k]) \mathbf{s}_{i,abc}[k] \quad (13b)$$

$$\bar{\mathbf{u}}_{dq}[l|k] = \frac{T_s - T_i}{T_s} \mathbf{u}_{dq}[k] + \frac{T_i}{T_s} \mathbf{u}_{i,dq}[k] \quad (13c)$$

$$\mathbf{i}_{dq}[k+1] = \mathbf{A}[k] \mathbf{i}_{dq}[k] + \mathbf{B}[k] \bar{\mathbf{u}}_{dq}[k] + \mathbf{E}[k]. \quad (13d)$$

In (13c), the voltages of the intermediate and the switching vectors are averaged. Because of the finite set of the switching states, the cost function can be evaluated for all feasible switching states $\mathbf{s}_{abc}^T[k]$. The switching state, from which a minimum cost results, is denoted as $\mathbf{s}_{abc}^*[k]$. This switching state is applied and the procedure must be repeated at the next sampling instant.

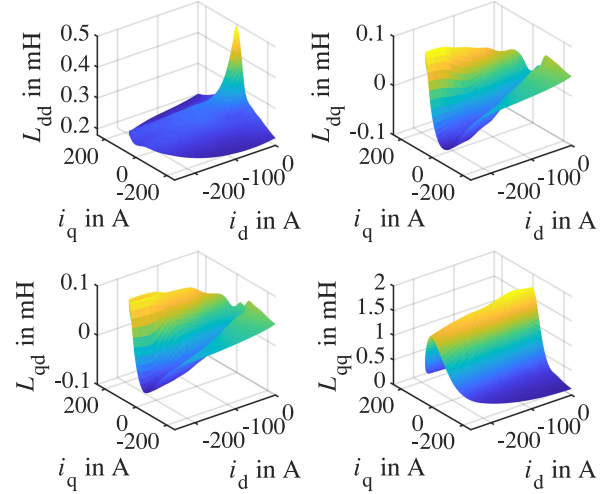


Fig. 3. Differential inductances of the highly utilized test motor.

III. EXPERIMENTAL TEST SETUP

All following experimental results for this contribution have been obtained on a laboratory test bench. The electrical drive system under test is a highly utilized interior permanent magnet synchronous motor (Brusa: HSM1-6.17.12-C01) for automotive applications and a two-level IGBT inverter (Semikron: 3×SKiiP 1242GB120-4D). For illustration purpose, the differential inductances are shown in Fig. 3, which highlights the significant (cross-)saturation present in the device under test. Additionally, a speed-controlled induction machine as load motor (Schorch: LU8250M-AZ83Z-Z) is mechanically coupled with the test motor. The testbench is further equipped with a dSPACE DS1006MC rapid-control-prototyping system. All measurements have been obtained by the dSPACE analog-digital-converters that have been synchronized with the control task. The most important test motor, inverter, control, and test bench parameters are listed in Table I. Due to symmetry conditions, the inductances L_{dq} and L_{qd} must be identical and for the operation point $\mathbf{i}_{dq} = \mathbf{0}$ A even zero. Because of measurement inaccuracies, deviations in the range of a few μH can be detected. In addition, Fig. 4 shows the test bench with a transient recorder in the foreground and the test motor in the background.

IV. OFFLINE ORDINARY LEAST SQUARES (OLS)

Before various models for online system identification are tested in a closed control loop, a dataset was recorded for offline preinvestigations. This dataset is available as supplementary material [34]. The FCS-MPC (13) was used as control algorithm for recording the dataset, whereby (4) was applied as prediction model. Here, the entries of the differential inductance matrix $\mathbf{L}_{dq,\Delta}$ and the flux vector $\boldsymbol{\psi}_{dq}$ were continuously updated with the use of LUTs in dependence on $\mathbf{i}_{dq}$. The dataset was recorded at the rated current operating point $\mathbf{i}_{dq}^* = [-170 \text{ A} \ 170 \text{ A}]^T$ at 2000 min^{-1} and contains 40000 samples that corresponds to a measuring time of 2 s. The quality of different models was investigated offline on the basis of the dataset using the OLS

TABLE I
PMSM, INVERTER, CONTROL, AND TEST BENCH PARAMETERS

PMSM	Brusa HSM16.17.12-C01	
Nominal power	P_{mech}	55 kW
Pole pair number	p	3
DC-link voltage	u_{DC}	300 V
Nominal current	I	170 A
Stator resistance	R_s	18 mΩ
Permanent magnet flux ($i_{\text{dq}} = 0$)	ψ_p	68 mVs
Diff. inductances ($i_{\text{dq}} = 0$)	L_{dd}	0.49 mH
	L_{dq}	6.0 μH
	L_{qd}	36 μH
	L_{qq}	1.7 mH
Inverter	3×SKiiP 1242GB120-4D	
Typology	voltage source inverter	
	2-level, IGBT	
Inverter interlocking time	T_i	3.3 μs
MPC settings		
Controller cycle time	T_s	50 μs
MPC prediction horizon	N	1
Controller hardware	dSPACE	
Processor board	DS1006MC, 4 cores, 2.8 GHz	
Measurement devices		
Oscilloscope	Tektronix MSO58	
Power analyzer	Yokogawa WT3000	
Current probes (zero-flux transducers)	3×Yokogawa, 500 A, 2 MHz	
Torque sensor	HBM, T10FS, 2 kN m	

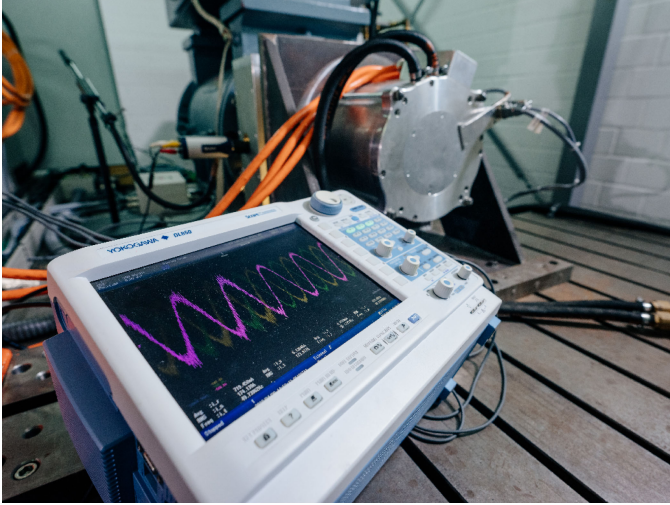


Fig. 4. Test bench with the investigated PMSM in the background.

method. Here, the quadratic deviation between measurement and model

$$\mathbf{r} = \mathbf{\Psi} - \mathbf{\Xi}\boldsymbol{\theta} \quad (14)$$

is minimized. Above, $\mathbf{\Psi}$ denotes the measurement vector, $\mathbf{\Xi}$ the regressor matrix and $\boldsymbol{\theta}$ the parameter vector. The squared Euclidean norm of the residuals $J(\boldsymbol{\theta}) = \|\mathbf{r}\|_2^2 = (\mathbf{\Psi} - \mathbf{\Xi}\boldsymbol{\theta})^T(\mathbf{\Psi} - \mathbf{\Xi}\boldsymbol{\theta})$ acts as cost function. The optimal parameter vector $\boldsymbol{\theta}^*$ can then be calculated with the following equation [35]:

$$\boldsymbol{\theta}^* = (\mathbf{\Xi}^T \mathbf{\Xi})^{-1} \mathbf{\Xi}^T \mathbf{\Psi} = \mathbf{\Xi}^\dagger \mathbf{\Psi}. \quad (15)$$

To identify the entries of the system matrices $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{E}$, state-space representation (6) can be rewritten into two separate

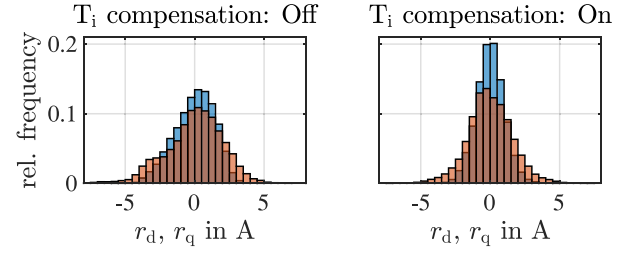


Fig. 5. Histogram of the residuals in r_d (orange) and r_q (blue) without (left) and with (right) interlocking time compensation.

LS problems.

$$\begin{aligned} & \underbrace{\begin{bmatrix} i_d[2] \\ \vdots \\ i_d[M+1] \end{bmatrix}}_{\mathbf{\Psi}_d} \\ &= \underbrace{\begin{bmatrix} i_d[1] & i_q[1] & \bar{u}_d[1] & \bar{u}_q[1] & 1 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ i_d[M] & i_q[M] & \bar{u}_d[M] & \bar{u}_q[M] & 1 \end{bmatrix}}_{\mathbf{\Xi}_d} \underbrace{\begin{bmatrix} a_{11} \\ a_{12} \\ b_{11} \\ b_{12} \\ e_1 \end{bmatrix}}_{\boldsymbol{\theta}_d} + \underbrace{\begin{bmatrix} r_d[1] \\ \vdots \\ r_d[M] \end{bmatrix}}_{\mathbf{r}_d} \end{aligned} \quad (16)$$

and

$$\underbrace{\begin{bmatrix} i_q[2] \\ \vdots \\ i_q[M+1] \end{bmatrix}}_{\mathbf{\Psi}_q} = \underbrace{\begin{bmatrix} a_{21} \\ a_{22} \\ b_{21} \\ b_{22} \\ e_2 \end{bmatrix}}_{\boldsymbol{\theta}_q} + \underbrace{\begin{bmatrix} r_q[1] \\ \vdots \\ r_q[M] \end{bmatrix}}_{\mathbf{r}_q} \quad (17)$$

with $\mathbf{\Xi}_q = \mathbf{\Xi}_d$ and $M \in \mathbb{N}$. The constant e_1 can be estimated with the help of the pseudoregressor 1 in the regression matrix. Having calculated the parameter vectors $\boldsymbol{\theta}_d^*$ and $\boldsymbol{\theta}_q^*$, the residuals r_d and r_q can be computed with (14).

A. Residual Analysis

Residual analysis plays an important role in validating the regression model. There are many statistical tools for model validation, but the primary tool for most modeling applications is graphical residual analysis [36]. Often zero mean and normally distributed noise is assumed for the residuals. The histogram plot of the residuals can be used to check whether the residuals are normally distributed. In Fig. 5, the residuals with and without interlocking time compensation can be seen. With interlocking time compensation, the variance can be reduced and also the symmetry of the distribution is improved. In electrical drives, there are also harmonics due to the motor geometry. For example, nonideal winding schemes or asymmetries result in a nonideal sinusoidal flux distribution in the air gap. The fundamental

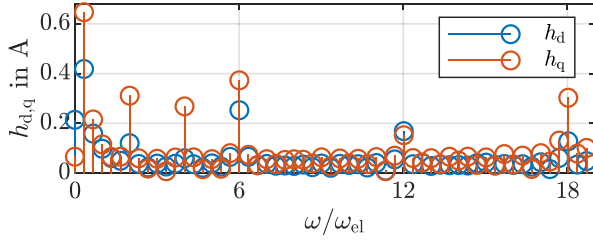


Fig. 6. Fourier coefficients h_d and h_q of the residuals r_d and r_q over electrical orders with active interlocking time compensation.

wave models (16) and (17) are not able to predict these current harmonics. For this reason, a discrete Fourier transform of the residuals is shown in Fig. 6. Here, the characteristic 6th, 12th, and 18th harmonics in the dq coordinate system due to the winding scheme as well as the 3th subharmonic can be observed. Due to the pole pair number of 3, this subharmonic corresponds to the fundamental frequency of the mechanical system. One reason for this subharmonic can be asymmetries between the pole pair systems.

V. REGRESSION MODELS

In the following, different model structures are defined to compare the performance in the offline OLS method and later also in the online RLS method.

1) *DFW Model*: The DFW model is able to map the behavior of the generalized motor model (4) with dense system matrices and (cross-)saturation effects. The m th line of the regressor matrices $\Xi_{d,DFW}$ and $\Xi_{q,DFW}$ are defined as in (16) and (17)

$$\xi_{d,DFW}[m]^T = [i_d[m] \quad i_q[m] \quad u_d[m] \quad u_q[m] \quad 1] \quad (18)$$

$$\xi_{q,DFW}[m]^T = \xi_{d,DFW}[m]^T. \quad (19)$$

The parameter vectors are

$$\theta_{d,DFW} = [a_{11} \quad a_{12} \quad b_{11} \quad b_{12} \quad e_1]^T \quad (20)$$

$$\theta_{q,DFW} = [a_{21} \quad a_{22} \quad b_{21} \quad b_{22} \quad e_2]^T. \quad (21)$$

For ease of reading, the index m will be omitted in the remainder of this work.

2) *Sparse Fundamental-Wave Model*: In comparison to the DFW model, the parameters b_{12} , b_{21} , and e_1 are set to zero for the sparse fundamental-wave (SFW) model. Thus, this model structure can only represent a model with sparse system matrices B and E . For this reason, no cross-saturation effects can be modeled [see Section II-B]. The m th line of the regressor matrices $\Xi_{d,SFW}$ and $\Xi_{q,SFW}$ are defined as

$$\xi_{d,SFW}^T = [i_d \quad i_q \quad u_d], \quad \xi_{q,SFW}^T = [i_d \quad i_q \quad u_q \quad 1]. \quad (22)$$

The parameter vectors are

$$\theta_{d,SFW} = [a_{11} \quad a_{12} \quad b_{11}]^T \quad (23)$$

$$\theta_{q,SFW} = [a_{21} \quad a_{22} \quad b_{22} \quad e_2]^T. \quad (24)$$

3) *Augmented Fundamental-Wave Model*: As seen in Section IV-A, the residuals still contain deterministic components in the form of harmonics. In order to identify a model that can represent harmonics, sine and cosine terms must be added, in a similar way as in [37], to the regressor matrix with the corresponding frequency of the harmonics. The number of parameters n to be estimated increase with each additional harmonic. In order to retain the real-time capability of the online system identification method, it is important to find a tradeoff between a limited number of parameters and model accuracy. A suitable method for parameter shrinkage and selection is LASSO regression [38]. For the LASSO method, regressors of the DFW model (18) and (19), as well as these multiplied by sine and cosine terms of order z , were applied

$$\xi_{d,LASSO}^T = [\xi_{d,DFW}^T \quad \sin(z\varepsilon)\xi_{d,DFW}^T \quad \cos(z\varepsilon)\xi_{d,DFW}^T] \quad (25)$$

$$\xi_{q,LASSO}^T = [\xi_{q,DFW}^T \quad \sin(z\varepsilon)\xi_{q,DFW}^T \quad \cos(z\varepsilon)\xi_{q,DFW}^T]. \quad (26)$$

The most important regressors can now be selected using LASSO regression. This was repeated for different dominant orders [see Fig. 6]. After selection of the most important regressors the m th line of the regressor matrices $\Xi_{d,AFW}$ and $\Xi_{q,AFW}$ can be defined as

$$\xi_{d,AFW} = \begin{bmatrix} i_d \\ i_q \\ u_d \\ u_d \sin(\frac{1}{3}\varepsilon) \\ u_d \cos(\frac{1}{3}\varepsilon) \\ u_d \sin(6\varepsilon) \\ u_d \cos(6\varepsilon) \end{bmatrix}, \quad \xi_{q,AFW} = \begin{bmatrix} i_d \\ i_q \\ u_q \\ 1 \\ u_q \sin(\frac{1}{3}\varepsilon) \\ u_q \cos(\frac{1}{3}\varepsilon) \\ u_q \sin(6\varepsilon) \\ u_q \cos(6\varepsilon) \end{bmatrix}. \quad (27)$$

The parameter vectors are

$$\theta_{d,AFW} = [a_{11} \quad a_{12} \quad \tilde{b}_{11,0} \quad \tilde{b}_{11,1} \quad \tilde{b}_{11,2} \quad \tilde{b}_{11,3} \quad \tilde{b}_{11,4}]^T \quad (28)$$

$$\theta_{q,AFW} = [a_{21} \quad a_{22} \quad \tilde{b}_{22,0} \quad e_2 \quad \tilde{b}_{22,1} \quad \tilde{b}_{22,2} \quad \tilde{b}_{22,3} \quad \tilde{b}_{22,4}]^T. \quad (29)$$

The matrix elements of the system matrix B are time-variant now. Interestingly, although a highly utilized drive with significant saturation and cross-saturation effects is used, the parameters b_{12} , b_{21} , and e_1 are set to zero for this operating point with LASSO regression. This is an indication that in this operating point modeling of cross saturation has a smaller influence on prediction quality than modeling of harmonics. A big disadvantage of this model, however, is that there is not enough excitation to identify the parameters for the sine and cosine regressors for low speeds. If the speed declines to standstill, the regressor matrix becomes rank-deficient and, therefore, the parameters cannot be uniquely determined.

TABLE II
OLS PERFORMANCE COMPARISON AT RATED MOTOR OPERATION

Model	DFW		SFW		AFW	
T _i comp.	Off	On	Off	On	Off	On
d-Axis						
$\mu(r_d)$ in mA	0.4	0.0	20.2	9.9	20.8	10.1
$\sigma(r_d)$ in A	1.443	1.021	1.572	1.064	1.414	0.937
R_d^2	0.986	0.993	0.940	0.971	0.941	0.972
q-Axis						
$\mu(r_q)$ in mA	0.0	-0.3	-0.2	0.0	0.0	0.0
$\sigma(r_q)$ in A	2.043	1.598	2.053	1.601	1.865	1.364
R_q^2	0.971	0.982	0.970	0.982	0.976	0.987

A. Performance Comparison Utilizing an Offline-OLS

In order to compare the quality of the different models and the effect of the interlocking time compensation, the residuals for the d - and q -axis with and without interlocking time compensation were calculated using the dataset of Section IV [34]. Performance criteria are the mean $\mu(r_{d,q})$ and the standard deviation $\sigma(r_{d,q})$ of the residuals. Additionally the coefficient of determination $R_{d,q}^2$ is evaluated. The values for these criteria are included in Table II. It can be seen that for every model the interlocking time compensation can improve the values for every performance criteria. Even though the mean values of the residuals in the d -axis for the SWF and AFW models are higher than that of the DFW model, these deviations are in the range of a few milliamperes, and, therefore can be considered as negligible. In consequence, it can be said that the residuals of all models are bias-free. The coefficient of determination for every model with interlocking time compensation is larger than 0.97. This indicates a sufficiently good model quality. Especially, $R_d^2 = 0.993$ for the DFW model with interlocking time compensation in the d -axis is excellent.

VI. RECURSIVE LEAST SQUARES

The OLS method was used to compare different models on the basis of a dataset offline. In contrast to the offline OLS method, the RLS method is now used for online system parameter identification. The model parameters are recursively corrected with each new measurement. In addition, past measurements can be weighted lower so that recent measurements are more relevant for parameter estimation. The calculations that must be executed in each iteration can be formulated as follows [39]:

$$\gamma[k] = \frac{\mathbf{P}[k]\boldsymbol{\xi}[k+1]}{\lambda + \boldsymbol{\xi}^T[k+1]\mathbf{P}[k]\boldsymbol{\xi}[k+1]} \quad (30)$$

$$\hat{\boldsymbol{\theta}}[k+1] = \hat{\boldsymbol{\theta}}[k] + \gamma[k] \left(\psi[k+1] - \boldsymbol{\xi}[k+1]^T \hat{\boldsymbol{\theta}}[k] \right) \quad (31)$$

$$\mathbf{P}[k+1] = \frac{1}{\lambda} (\mathbf{I} - \gamma[k]\boldsymbol{\xi}^T[k+1]) \mathbf{P}[k]. \quad (32)$$

where $\mathbf{P}$ is a matrix proportional to the covariance matrix $\text{Cov}(\hat{\boldsymbol{\theta}}, \hat{\boldsymbol{\theta}})$ of the estimated parameter vector and the forgetting factor, and λ is a weighting factor that assigns a lower weighting to the past measurements $\boldsymbol{\xi}$ of the regressors. In (30), a correction

term is calculated with which the parameter vector $\hat{\boldsymbol{\theta}}$ and matrix $\mathbf{P}$ are updated. The forgetting factor $0 < \lambda < 1$ acts as tuning parameter. An initial parameter vector $\hat{\boldsymbol{\theta}}[0]$ and an initial matrix $\mathbf{P}[0]$ also must be specified. In some operating points mainly a zero voltage vector is actuated by the MPC (e.g., very low speeds). If the RLS algorithm is updated at every sampling instant, this can lead to a rank deficient matrix $\mathbf{P}$ due to a lacking system excitation. For this reason, the RLS algorithm is only updating up to one zero voltage vector after each nonzero vector in order to limit the amount of consequent zero voltage vectors utilized.

VII. EXPERIMENTAL RESULTS

In order to test the accuracy of the system identification using the RLS method, the behavior in stationary and transient operation was evaluated. The system matrices identified online were used in the closed control loop to compensate the computation time and control delay and also as prediction model for the FCS-MPC algorithm. For the RLS method the initial parameter vectors $\hat{\boldsymbol{\theta}}_{d,q}[0]$ can be computed either with data sheet parameters (if available) or nameplate information in addition with simplifying assumptions. The forgetting factors were set to $\lambda_{d,q} = 0.99$ for all models. The matrices $\mathbf{P}_{d,q}[0]$ were selected as unit matrices. In the following, the term residuals refers to the one-step prediction error using the data-driven RLS models.

A. Steady-State Behavior

In order to gain a comprehensive impression of the quality of the models in stationary operation, 83 equispaced operating points were recorded in the left i_d - i_q half-plane with a maximum length $\|\mathbf{i}_{dq}\|$ of 250 A. Every dataset of one operating point contains 40 000 samples, which corresponds to a measuring time of 2 s at a constant speed of 2000 min⁻¹. The regression models with and without interlocking time compensation and also a LUT-based FCS-MPC as in Section IV are compared under different performance criteria. The LUT-based model is called WB model. In the following, performance criteria are defined, which are evaluated for each operating point. The length of the mean vector of the residuals is defined as

$$\|\boldsymbol{\mu}(\mathbf{r}_{dq})\| = \sqrt{(\mu(r_d))^2 + (\mu(r_q))^2} \quad (33)$$

to obtain a scalar quantity that gives an indication of the residuals' mean. To measure the standard deviation of the residuals, a vector is defined that contains the standard deviation of both residuals $\boldsymbol{\sigma}(\mathbf{r}_{dq}) = [\sigma(r_d) \quad \sigma(r_q)]^T$. The length of the vector $\|\boldsymbol{\sigma}\|$ is used to obtain a scalar value here as well. To evaluate the steady-state accuracy of the current controller in dependence of the models, the length of the steady-state control deviation vector is defined as

$$\|\mathbf{e}_{dq}\| = \|\boldsymbol{\mu}(\mathbf{i}_{dq}) - \mathbf{i}_{dq}^*\|. \quad (34)$$

A further requirement in steady-state operation is to decrease the harmonic current distortion in order to reduce ohmic as well as thermal losses. A suitable measure for the harmonic current

TABLE III
FCS-MPC PERFORMANCE COMPARISON FOR DIFFERENT MODELS IN STEADY-STATE AT 2000 min^{-1} IN THE ENTIRE LEFT $i_d - i_q$ HALF-PLANE

Model	DFW		SFW		AFW		WB		Koopman [27]
T_i comp.	✗	✓	✗	✓	✗	✓	✗	✓	✗
RLS		✓		✓		✓		✗	-
LUT		✗		✗		✗		✓	-
$\ \mu(r_{dq})\ $ in A	0.007	0.046	0.048	0.170	0.043	0.168	2.372	1.445	0.187
$\ \sigma(r_{dq})\ $ in A	1.946	1.533	2.765	2.440	2.714	2.381	4.951	4.942	1.870
$\ e_{dq}\ $ in A	0.850	0.798	1.063	1.037	0.978	1.010	2.931	4.566	0.963
I_{TDD} in %	5.75	5.71	5.83	5.79	5.89	5.82	5.99	6.02	6.32
Turnaround time in μs	30.9		29.5		32.9		29.4		28.2

distortion is the total demand distortion (TDD)

$$I_{TDD} = \frac{\sqrt{\sum_{h \neq 1} I_h^2}}{I_{nom}} \quad (35)$$

in which the nominal current $I_{nom} = \frac{1}{\sqrt{2}} \|i_{dq}\| = 170 \text{ A}$ refers to the operating condition at $i_{dq} = [-170 \text{ A} \quad 170 \text{ A}]^T$. The harmonic components I_h , can be differentiated into the fundamental component I_1 and the h th harmonic and subharmonic components I_h in the abc coordinate system. The total harmonic distortion (THD) refers to the actual current rather than the nominal current. For this reason, the THD tends to go to infinity for small fundamental components while the TDD remains effectively constant. Therefore, the TDD is a more suitable choice for measuring the current distortion than the THD [40]. In order to obtain scalar values of the performance criteria for the complete left $i_d - i_q$ half-plane, these are averaged over all $L = 83$ operating points

$$\bar{x} = \frac{1}{L} \sum_{l=1}^L x_l. \quad (36)$$

Table III compiles the resulting performance criteria. The residuals of all regression models have a lower mean and standard deviation compared to the residuals of the WB model. Due to the increased prediction accuracy, the regression models achieve an improved control performance compared to the WB model, which results in a lower steady-state control deviation and lower current distortion. Although the mean value of the residuals without interlocking time compensation is lower, this is not a major disadvantage for the models with interlocking time compensation, since they are still in the order of some 100 mA. For the models with interlocking time compensation, there are lower values in the standard deviation and also in the current distortion. The average turnaround time increases only slightly with increasing complexity of the RLS methods. No significant difference in the turnaround time with and without interlocking time compensation could be determined. This is why only one value per model is presented here. It is worth mentioning that the turnaround time for the SFW model is insignificantly higher than that of the WB model with LUTs. The mean and standard deviation of the residuals for the DFW model with interlocking time compensation can be seen in Fig. 7 and the current distortion in Fig. 8. Furthermore, a comparison was made to the method

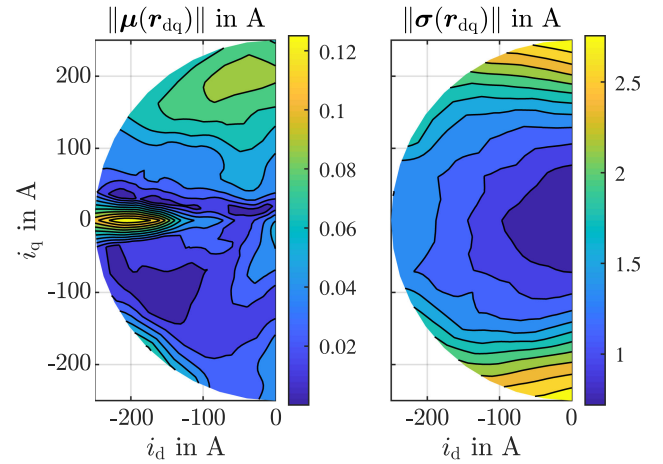


Fig. 7. Length of the mean $\|\mu\|$ and standard deviation vector $\|\sigma\|$ of the residuals r_d, r_q in the left $i_d - i_q$ half-plane for the DFW model with interlocking time compensation.

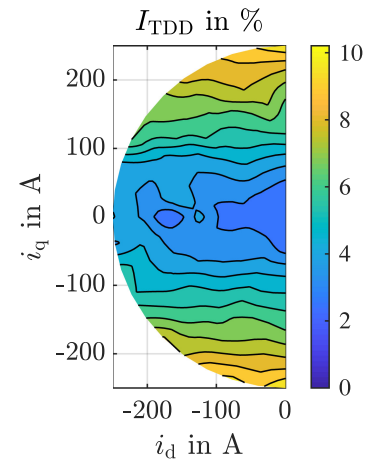


Fig. 8. TDD of the phase currents in the left $i_d - i_q$ half-plane for the DFW model with interlocking time compensation.

based on the Koopman operator from [27]. Here, an autonomous system for current prediction using a LS method was identified for each switching vector. The estimation performance is similar to that of the DFW model without interlocking time compensation. Moreover, it should be noted that the online calculation

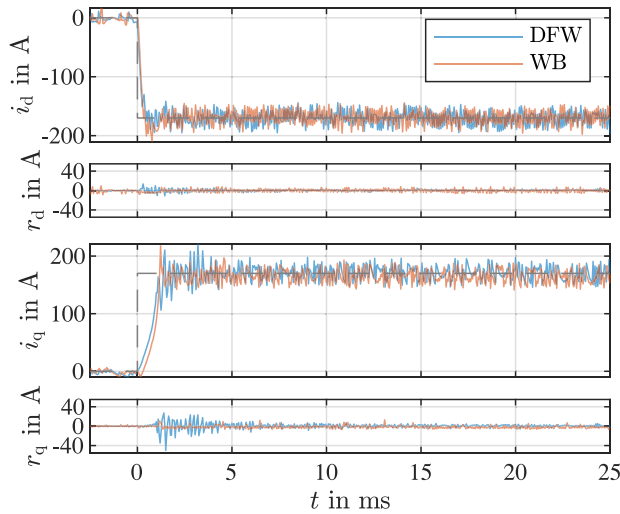


Fig. 9. Current step response at constant speed of 2000 min^{-1} . For the DFW as well as for the WB model, interlocking time compensation was used.

time of the Koopman-based control approach cannot be directly compared to the methods discussed in this contribution. The reasons for that are different control framework implementations leading to varying overhead calculation demand as well as that [27] is not running an online RLS with continuous system corrections but occasional LS-based system updates in a background task. Nevertheless, the Koopman-based MPC turnaround time is depicted for the sake of completeness indicating that the computational demand is in the same order of magnitude as the proposed identification methods in this article.

B. Transient Behavior

The RLS method identifies the system behavior at the present operating point. The identified system matrices are only valid locally and not globally for all operation points. For this reason, the behavior of the RLS method in transient operation with changing operating points is studied. Exemplarily, the DFW model with interlocking time compensation is used here. One reason for time-invariant system matrices are changing parameters such as differential and absolute inductances. These inductances have a strong dependency on the current vector i_{dq} , especially in the case of highly utilized motors. For this reason, the first scenario to be used is a step response, as shown in Fig. 9. The short settling time and the overshoot-free behavior of the FCS-MPC can be seen here. During the transient process, the residuals are slightly elevated but decrease toward their steady-state values within a few milliseconds. Nevertheless, the transient control performance during that reference step change is only slightly worse using RLS model adaption compared to the WB model. Not only the electrical parameters can change, but, also the frequency of the electrical system ω , which appears as a physical parameter in the system matrices. To study the influence of a changing frequency, the behavior of the current controller is investigated during speed transients, controlled by the load machine, as shown in Fig. 10. Having reached the constant cutoff rotational speed, the DFW models' absolute residuals likewise

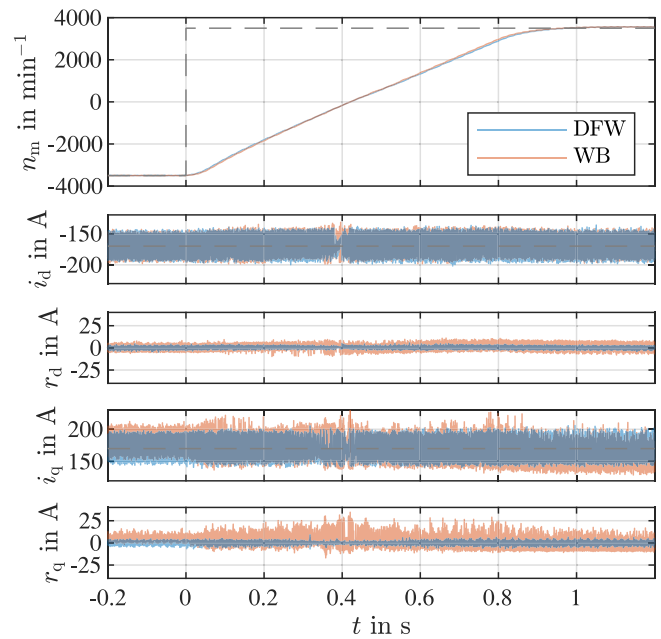


Fig. 10. Speed step response including speed-reversal at rated current. For the DFW as well as for the WB model, interlocking time compensation was used.

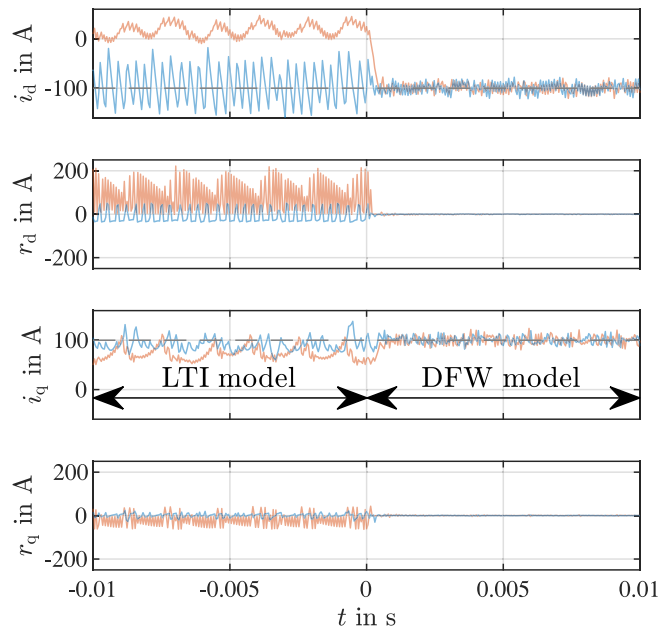


Fig. 11. Initialization of the DFW model at $t = 0 \text{ s}$ for drastically distorted starting parameters: red signals relate to 10% and blue signals to 1000% of the nominal motor parameters.

remain constant, highlighting their resilience to speed changes. The reason for this is the large time constant of the mechanical system compared to the fast adaptation of the RLS method in the electrical domain.

As mentioned in Section VI, the parameters of the RLS algorithm and, thus, the system matrices must be initialized. To investigate the robustness of the proposed algorithm during initialization, the system matrices of the DFW model were

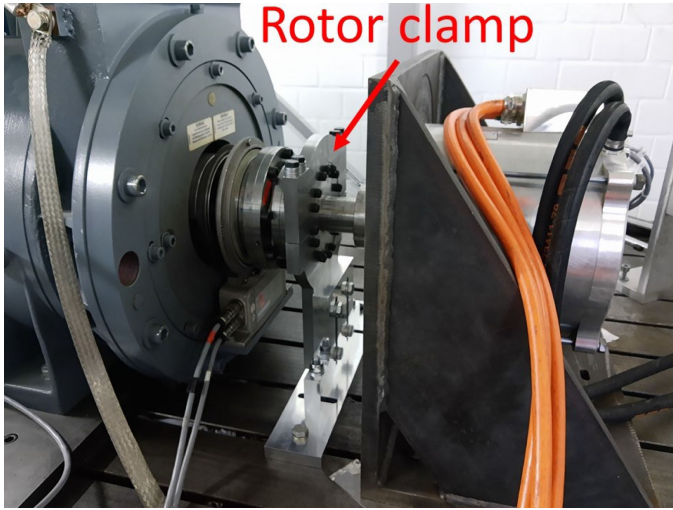


Fig. 12. Load and test motor with the used rotor clamp for standstill investigations

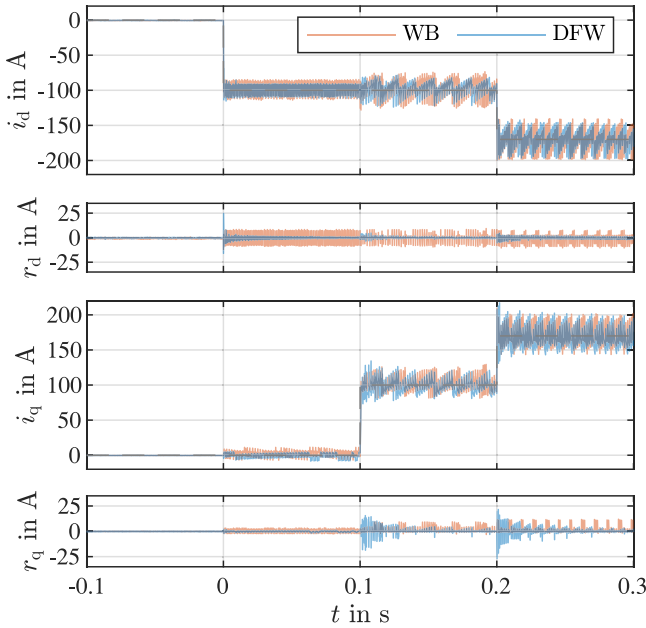


Fig. 13. Sequence of current step responses at standstill. For the DFW as well as for the WB model, interlocking time compensation was used.

initialized for two sets of drastically distorted motor parameters. Both sets are based on the motor parameters listed in [41]. Here, just a strongly simplified LTI motor model with a set of constant motor parameters L_d , L_q , R_s , and ψ_p was assumed. One set contains underrated motor parameter values that corresponds to just 10% and the other set contains overrated values that corresponds to 1000% of the nominal motor parameters. In Fig. 11, the initialization behavior at an operation point of $i_{dq} = [-100 \text{ A} \ 100 \text{ A}]^T$ with a speed of 2000 min^{-1} for both parameter sets is shown. Until $t = 0 \text{ s}$ the RLS algorithm is deactivated and the system matrices are based on the simplified motor model with the distorted parameter sets. Here, the worse

control performance can be seen. At $t = 0 \text{ s}$, the RLS algorithm is activated. The RLS algorithm converges in the range of a millisecond and demonstrates its robustness even when facing extremely distorted starting conditions..

Many online identification methods fail at standstill operation due to missing induced voltage feedback. In order to show the working principle and robustness of the proposed method at standstill, the rotor of the PMSM was mechanically blocked [see Fig. 12]. For testing reasons, some arbitrary current reference step changes were applied as shown in Fig. 13. Here, a behavior similar to the one from the current step response in Fig. 9 can be observed. During the transient processes, the residuals are slightly elevated but decrease towards their steady-state values within a few milliseconds. Therefore, the proposed data-driven RLS estimator is fully feasible even at standstill operation, which is an important feature e.g., for automated self-commissioning routines which are often executed without a rotating motor.

VIII. CONCLUSION

In this article, it has been shown that interlocking time compensation in combination with a data-driven RLS model identification improves the prediction accuracy and, therefore, the performance of the FCS-MPC significantly compared to a baseline WB model, which addresses varying parameter by offline identified LUTs. Different data-driven model topologies have been compared against each other in terms of computational runtime and accuracy. Comprehensive experimental investigations proof the performance of the RLS-based FCS-MPC during steady-state and transients in the entire electrical and speed operation range. In particular, the presented approach is highly suitable for self-commissioning drive applications since accurate WB models are not available in this scenario.

In the future, it could be investigated how physical parameters can be extracted from the identified models in an appropriate way. These physical parameters could be used to calculate the maximum torque per current and maximum torque per voltage characteristics as well as the torque for a higher-level operating point control. Moreover, it could be beneficial to add a long-term memory model (e.g., artificial neural network or gaussian process) to support the local data-driven RLS approach during transients.

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